

LM09 Parametric and Non-Parametric Tests of Independence

1. Introduction.....2

2. Tests Concerning Correlation2

3. Tests of Independence Using Contingency Table Data5

Summary.....7

Required disclaimer: IFT is a CFA Institute Prep Provider. Only CFA Institute Prep Providers are permitted to make use of CFA Institute copyrighted materials which are the building blocks of the exam. We are also required to create / use updated materials every year and this is validated by CFA Institute. Our products and services substantially cover the relevant curriculum and exam and this is validated by CFA Institute. In our advertising, any statement about the numbers of questions in our products and services relates to unique, original, proprietary questions. CFA Institute Prep Providers are forbidden from including CFA Institute official mock exam questions or any questions other than the end of reading questions within their products and services.

CFA Institute does not endorse, promote, review or warrant the accuracy or quality of the product and services offered by IFT. CFA Institute®, CFA® and “Chartered Financial Analyst®” are trademarks owned by CFA Institute.

© Copyright CFA Institute

Version 1.0

1. Introduction

This learning module covers:

- Parametric and non-parametric tests of correlation
- Tests of independence based on contingency table data

2. Tests Concerning Correlation

The strength of linear relationship between two variables is assessed through correlation coefficient. The significance of a correlation coefficient is tested by using hypothesis tests concerning correlation.

The most common way of setting up hypotheses concerning correlation is to check if the population correlation is not equal to 0. In this case the null and alternate hypothesis will be (ρ represents the population correlation coefficient):

- $H_0: \rho = 0$
- $H_a: \rho \neq 0$

It is a two-tailed test.

We can also set up hypothesis to check if the population correlation is positive or negative. The null and alternate hypothesis in these cases will be:

One sided (right side): $H_0: \rho \leq 0$ versus $H_a: \rho > 0$

One sided (left side): $H_0: \rho \geq 0$ versus $H_a: \rho < 0$

Parametric Test of a Correlation

As long as the two variables are distributed normally, we can use sample correlation, r for our hypothesis testing. The formula for the t-test is

$$t = \frac{r \sqrt{n-2}}{\sqrt{1-r^2}}$$

where: $n - 2$ = degrees of freedom if H_0 is true.

The magnitude of r needed to reject the null hypothesis $H_0: \rho = 0$ decreases as sample size n increases due to the following:

- As n increases, the number of degrees of freedom increases and the absolute value of the critical value t_c decreases.
- As n increases, the absolute value of the numerator increases, leading to larger-magnitude t-values.

In other words, as n increases, the probability of Type-II error decreases, all else equal.

Example

The sample correlation between the oil prices and monthly returns of energy stocks in a Country A is 0.7986 for the period from January 2014 through December 2018. Can we reject a null hypothesis that the underlying or population correlation equals 0 at the 0.05 level of significance?

Solution:

$H_0: \rho = 0 \rightarrow$ true correlation in the population is 0.

$H_a: \rho \neq 0 \rightarrow$ correlation in the population is different from 0.

From January 2014 through December 2018, there are 60 months, so $n = 60$. We use the following statistic to test the above.

$$t = \frac{0.7986 \sqrt{60-2}}{\sqrt{1-0.7986^2}} = \frac{6.0820}{0.6019} = 10.1052$$

At the 0.05 significance level, the critical level for this test statistic is 2.00 ($n = 60$, degrees of freedom = 58). When the test statistic is either larger than 2.00 or smaller than -2.00, we can reject the hypothesis that the correlation in the population is 0. The test statistic is 10.1052, so we can reject the null hypothesis.

Non-Parametric Test of Correlation: The Spearman Rank Correlation Coefficient**Instructor's Note:**

Focus on the basics of this topic, the probability of being tested on the details is low.

If the two variables under consideration are not normally distributed, we can use a test based on the Spearman rank correlation coefficient, r_s . The Spearman rank correlation coefficient is equivalent to the usual correlation coefficient but is calculated on the *ranks* of two variables within their respective samples.

Steps for calculating r_s :

1. Sort the X observations from largest to smallest. Assign the number 1 to the largest value observation, the number 2 to the second largest value observation, and so on. In the event of a tie, assign the average of the ranks that the tied observations share to each tied observation. For example, if the third and fourth largest values are tied, we assign a rank of 3.5 to both observations (the average of 3 and 4). Repeat the procedure for the observations on Y.
2. Calculate the difference in ranks, d_i , for each pair of observations on X and Y, and then calculate d_i^2 (the squared difference in ranks).
3. For a sample size n , the spearman rank correlation is:

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

Example:

(This is based on Question Set from the curriculum.)

You want to determine whether the alpha of a mutual fund is correlated with its expense ratio. You gather the following information about 9 mutual funds.

Mutual Fund	Alpha	Expense Ratio
1	-0.52	1.34
2	-0.13	0.40
3	-0.50	1.90
4	-1.01	1.50
5	-0.26	1.35
6	-0.89	0.50
7	-0.42	1.00
8	-0.23	1.50
9	-0.60	1.45

Perform a Spearman rank correlation test based on this sample data. Determine whether to reject the null hypothesis at the 0.05 level of significance if the critical values are ± 2.306 .

Solution:

The null and alternate hypotheses are:

$H_0: \rho = 0$ versus $H_a: \rho \neq 0$

To calculate the Spearman rank correlation, we first construct the following table:

Mutual Fund	Alpha	Expense Ratio	Rank by Alpha	Rank by Expense Ratio	Difference in Rank	Difference Squared
1	-0.52	1.34	6	6	0	0
2	-0.13	0.40	1	9	-8	64
3	-0.50	1.90	5	1	4	16
4	-1.01	1.50	9	2.5	6.5	42.25
5	-0.26	1.35	3	5	-2	4
6	-0.89	0.50	8	8	0	0
7	-0.42	1.00	4	7	-3	9
8	-0.23	1.50	2	2.5	-0.5	0.25
9	-0.60	1.45	7	4	3	9
					Sum	144.5

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

$$r_s = 1 - \frac{6(144.5)}{9(9^2 - 1)} = -0.20416$$

The test statistic is calculated as:

$$t = \frac{r \sqrt{n-2}}{\sqrt{1-r^2}}$$

$$t = \frac{-0.20416 \sqrt{9-2}}{\sqrt{1-0.20416^2}} = -0.55177$$

Since this value falls within the range of ± 2.306 , we fail to reject the null hypothesis that the Spearman rank correlation coefficient is zero.

3. Tests of Independence Using Contingency Table Data

Instructor's Note:

Focus on the basics of this topic, the probability of being tested on the details is low.

When dealing with categorical or discrete data presented in the form of a contingency table, we use a chi-squared distributed test statistic.

Consider the following contingency table which classifies 1,594 ETFs based on two dimensions: size and investment type.

Investment Type	Size Based on Market Capitalization			
	Small	Medium	Large	Total
Value	50	110	343	503
Growth	42	122	202	366
Blend	56	149	520	725
Total	148	381	1,065	1,594

Suppose we want to test whether a relationship exists between the size and investment type, we can perform a test of independence using a chi-squared distributed test statistic. This nonparametric test compares actual observed frequencies with those expected on the basis of independence.

The test statistic is calculated as:

$$\chi^2 = \sum_{i=1}^m \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where:

m = number of cells in the table

O_{ij} = observed frequency

E_{ij} = expected frequency calculated as: $E_{ij} = \frac{(\text{Total row } i) \times (\text{Total column } j)}{\text{Overall total}}$

This test statistic has degrees of freedom of $(r-1)(c-1)$, where r is the number of categories for the first variable and c is the number of categories of the second variable.

The expected frequency for one cell – small cap value can be calculated as:

$$E_{ij} = \frac{503 \times 148}{1,594} = 46.703$$

In the same way we can calculate the expected frequency for all 9 cells to get the following table.

Expected Frequency of ETFs by Size and Investment Type

Investment Type	Size Based on Market Capitalization		
	Small	Medium	Large
Value	46.703	120.228	336.070
Growth	33.982	87.482	244.536
Blend	67.315	173.290	484.395
Total	148.000	381.000	1,065.000

We then calculate the squared difference between observed and expected frequencies scaled by the expected frequency for each cell.

$$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

For the small cap value cell, this calculation is:

$$\frac{(50 - 46.703)^2}{46.703} = 0.2327$$

We perform the same calculations for other cells and get the following table:

Scaled Squared Deviation for Each Combination of Size and Investment Type

Investment Type	Size Based on Market Capitalization		
	Small	Medium	Large
Value	0.233	0.870	0.143
Growth	1.892	13.620	7.399
Blend	1.902	3.405	2.617

Finally, we sum all the above values to get a chi-squared test statistic as 32.08025.

Testing Our Hypothesis

The null and alternate hypothesis are:

H_0 : ETF size and investment type are not related, so these classifications are independent;

H_a : ETF size and investment type are related, so these classifications are not independent.

With a $(3-1) \times (3-1) = 4$ degrees of freedom and a one-sided test with a 5% level of significance, the critical value is 9.4877.

Since the calculated chi-squared test statistic (32.08025) is greater than 9.4877 we reject the null hypothesis of independence and conclude that ETF size and investment type are related.

Summary

LO: Explain parametric and nonparametric tests of the hypothesis that the population correlation coefficient equals zero, and determine whether the hypothesis is rejected at a given level of significance.

Parametric test: The significance of a correlation coefficient is tested by using hypothesis tests concerning correlation. The formula for the t-test is:

$$t = \frac{r \sqrt{n - 2}}{\sqrt{1 - r^2}}$$

where: $n - 2$ = degrees of freedom

Non-parametric test: The Spearman rank correlation coefficient is equivalent to the usual correlation coefficient but is calculated on the *ranks* of two variables within their respective samples.

LO: Explain tests of independence based on contingency table data.

A chi-square distributed test statistic is used to test for independence of two categorical variables. This nonparametric test compares actual frequencies with those expected on the basis of independence.

This test statistic has degrees of freedom of $(r - 1)(c - 1)$, where r is the number of categories for the first variable and c is the number of categories of the second variable.